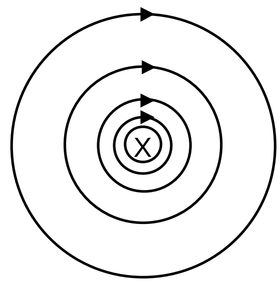
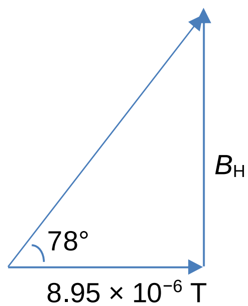
<table>
<tr><td></td><td>$I_B \quad (12)$</td></tr>
<tr><td>4(a)</td><td></td></tr>
<tr><td>4(b)</td><td>$B = \frac{\mu_0 I}{2\pi r}$
$= \frac{(4\pi \times 10^{-7})(8.5)}{2\pi(0.19)}$
$= 8.95 \times 10^{-6}$ T</td></tr>
<tr><td>4(c)(i)</td><td>

$\tan 78° = \frac{B_H}{8.95 \times 10^{-6}}$
$B_H = 4.21 \times 10^{-5}$ T</td></tr>
<tr><td>4(c)(ii)</td><td>The resultant magnetic flux density is zero at the point where the flux density due to the wire is directed downwards with a magnitude of 4.21 × 10^{-5} T so it exactly cancels the flux density due to Earth.

$4.21 \times 10^{-5} = \frac{\mu_0 (8.5)}{2\pi r}$
$r = 0.040$ m

This point is 0.040 m to the right of the wire.</td></tr>
</table>

5(a)(i)	[illegible]